

Supplement: Optimal Siting and Sizing of Virtual Power Lines in Medium-Voltage Grids for Congestion Management

Flin Verdaasdonk^{a,*}, Maria Vlasidou^a, Gerwin Hoogsteen^a, Johann Hurink^a

^a*Department of Applied Mathematics, University of Twente, Enschede, Netherlands*

1. Overview

This document contains supplementary results and algorithms used for the paper "Optimal Siting and Sizing of Virtual Power Lines in Medium-Voltage Grids for Congestion Management" submitted to EPSR in June 2026.

It is **not** a standalone document.

- **Appendix I** presents the results shown in the source paper, but before aggregation, i.e. per scenario.
- **Appendix II** contains an analyses of the solve time of the DC-OPF formulation under various conditions. Although exemplary of the computational cost for even this simple (i.e. MILP opposed to MISOCP) formulation, the results deviated from the main line of the paper.
- **Appendix III** contains a brief numerical validation of DC-OPF. We shown that DC-OPF, despite it limitations for MV-grids, optimizes towards sensible behavior, minimizing extreme currents and power flows.
- **Appendix IV** contains algorithms detailing how the SimBench grids were preprocessed.

For code and data, see https://github.com/Flinverdaasdonk/Optimal_Siting_and_Sizing_of_Virtual_Power_Lines.

I. Results per different grid groups and cable limits

Here we plot the results for our analyses on other grid groups and cable limits.

We have 12 possible scenarios (grid groups $\times$ cable limits). For each scenario, we compute the main results presented in the paper. Specifically:

- Fig. A.1 corresponds to Fig. 2 and shows the results comparing optimized-versus-random placement.
- Fig. A.2 corresponds to Fig. 3 and shows the importance of the total installed capacity.
- Fig. A.3 corresponds to Fig. 4a and shows the importance of the maximum number of Storage Devices (SDs).
- Fig. A.5 corresponds to Fig. 4b and shows the importance of maximum charge-rate, more specifically the C-rating ρ .
- Fig. A.4 corresponds to Fig. 4c and shows the importance of the maximum installed battery size.

II. Effects on Solve Time

Throughout our study we identified various factors that significantly increased the solve time of each experiment. We investigate the effect of the following parameters on the solve time:

- The size of the time set $|\mathcal{T}|$.
- The number of nodes $|\mathcal{N}|$.
- The size of the candidate set $|\mathcal{N}^A|$.
- The number of SDs Γ .

Unless swept over, the solve-time experiments use (i) a time set $|\mathcal{T}|$ of two weeks, (ii) a fixed number of SDs $\Gamma = 5$, (iii) a fixed candidate set with size $|\mathcal{N}^A| = 27$, and (iv) a fixed size $|\mathcal{N}| = 97$. The total installed capacity is 440 MWh and the specific MV grids in this experiment are 1-MV-comm-0-sw, 1-MV-rural-0-sw, and 1-MV-semi-urban-0-sw of SimBench.

II.1. Results of Study on Solve Time

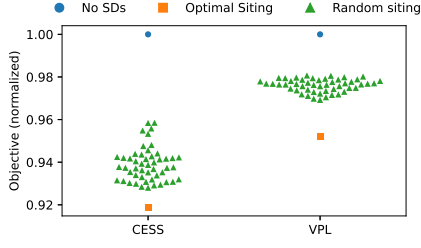
This section analyzes which parameters most impact the solve time of the MILP formulation. The considered parameters are the time set size $|\mathcal{T}|$, the number of nodes $|\mathcal{N}|$, the size of the candidate set $|\mathcal{N}^A|$, and the number of SDs Γ . The results are shown in log-scale in Fig. B.6.

*Corresponding author

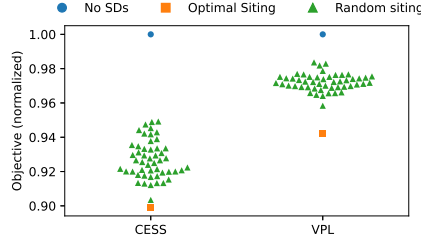
Email address: f.verdaasdonk@utwente.nl (Flin Verdaasdonk)

¹This publication is part of the project ROBUST: Trustworthy AI-based Systems for Sustainable Growth (KICH3.LTP.20.006), financed (partly) by the Dutch Research Council (NWO), Industry and the Dutch Ministry of Economic Affairs and Climate Policy under LTP KIC 2020-2023. All content represents the opinion of the authors, not necessarily shared or endorsed by their respective employers and/or sponsors.

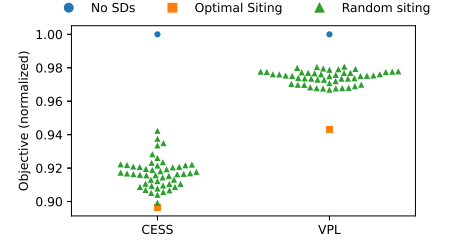
Optimal versus random placement



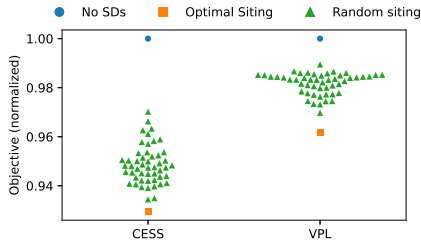
(a) Grid C0S1U2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



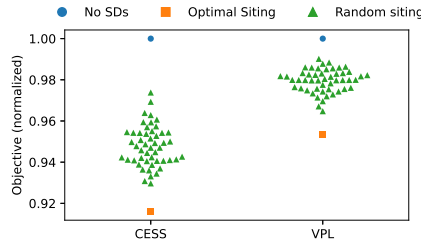
(b) Grid C1R0S2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



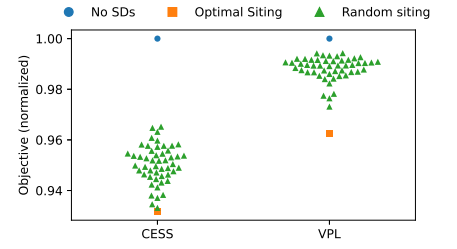
(c) Grid C2R1U0, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



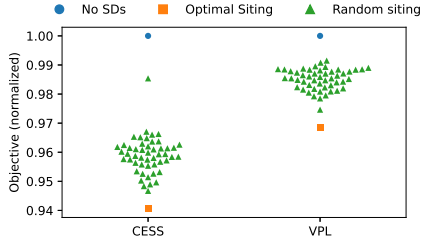
(d) Grid C0S1U2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



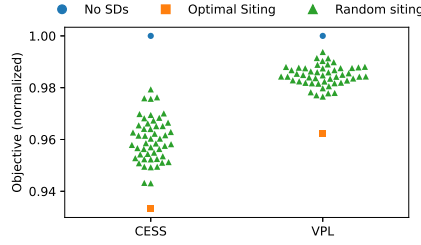
(e) Grid C1R0S2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



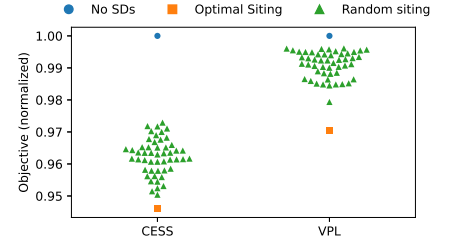
(f) Grid C2R1U0, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



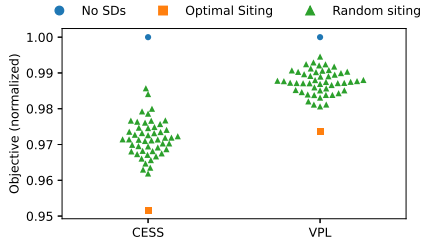
(g) Grid C0S1U2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



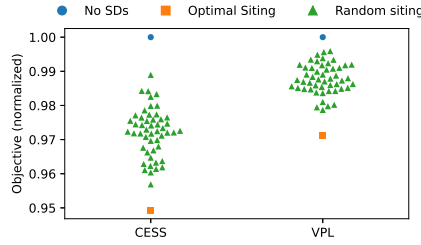
(h) Grid C1R0S2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



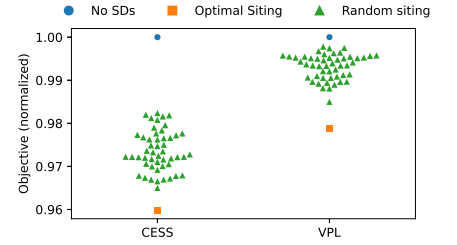
(i) Grid C2R1U0, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



(j) Grid C0S1U2, $\bar{P} = 0$, i.e. power flow minimization.



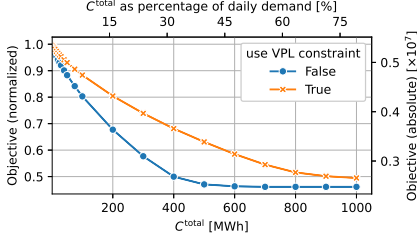
(k) Grid C1R0S2, $\bar{P} = 0$, i.e. power flow minimization.



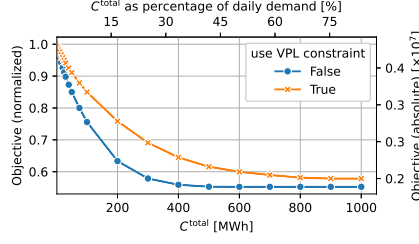
(l) Grid C2R1U0, $\bar{P} = 0$, i.e. power flow minimization.

Figure A.1: The effect of no placement, optimal siting, and a given random siting, for various grids and congestion definitions. All twelve panels show the same qualitative conclusion: optimal siting matters, and it matters more for Virtual Power Lines (VPLs) than for Conventional Energy Storage Systems. Random Conventional Energy Storage Systems often still perform reasonably, whereas random Virtual Power Lines (VPLs) sit much closer to the no-storage benchmark.

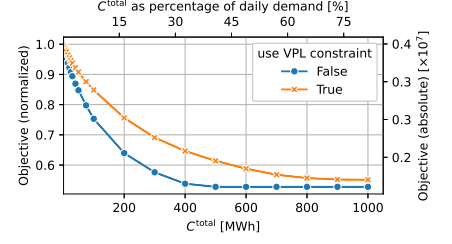
Effect of increasing total installed capacity



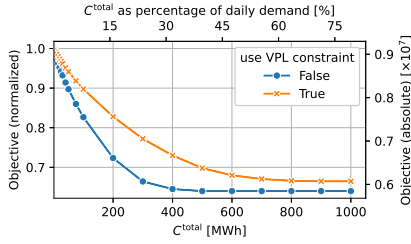
(a) Grid C0S1U2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



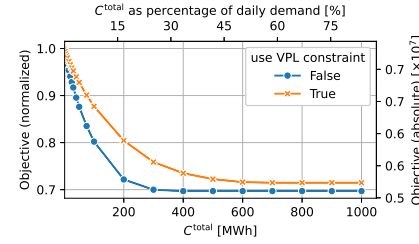
(b) Grid C1R0S2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



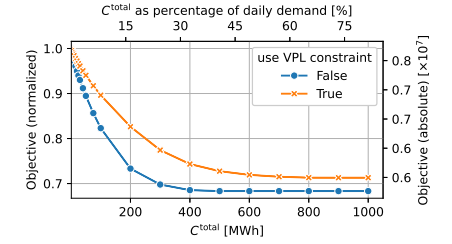
(c) Grid C2R1U0, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



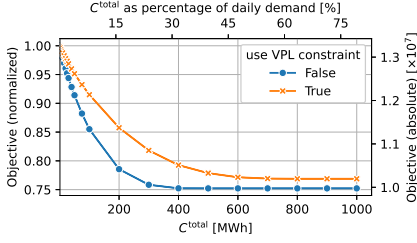
(d) Grid C0S1U2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



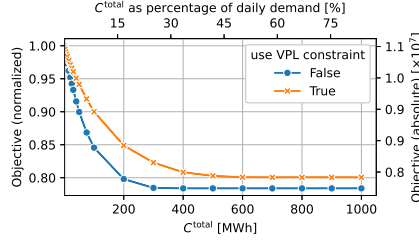
(e) Grid C1R0S2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



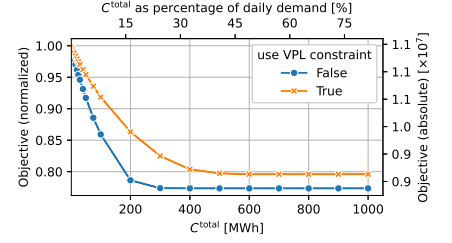
(f) Grid C2R1U0, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



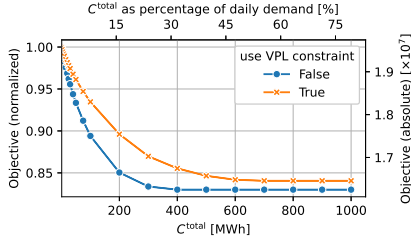
(g) Grid C0S1U2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



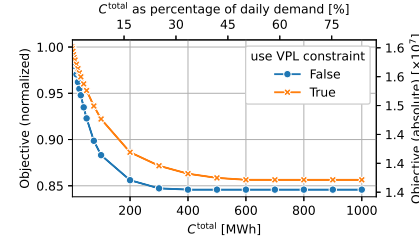
(h) Grid C1R0S2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



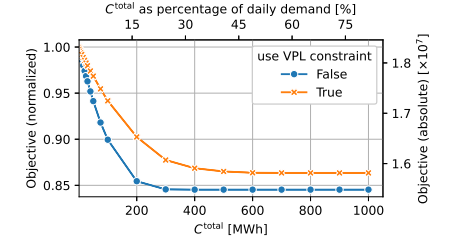
(i) Grid C2R1U0, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



(j) Grid C0S1U2, $\bar{P} = 0$, i.e. power flow minimization.



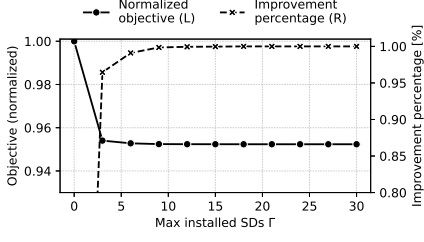
(k) Grid C1R0S2, $\bar{P} = 0$, i.e. power flow minimization.



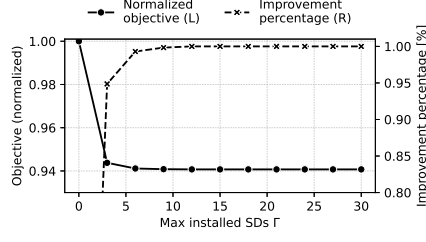
(l) Grid C2R1U0, $\bar{P} = 0$, i.e. power flow minimization.

Figure A.2: The effect of total installed capacity for both Virtual Power Lines (VPLs) and Conventional Energy Storage Systems, for various grids (columns) and congestion definitions (rows). Daily demand computed based $\sum |p|$ of all nodes in the composed grid. Overall we see the same trend. Virtual Power Lines (VPLs) approach Conventional Energy Storage Systems but never obtain the same improvement. Both Virtual Power Lines (VPLs) and Conventional Energy Storage Systems saturate in possible reduction for large values of installed capacity.

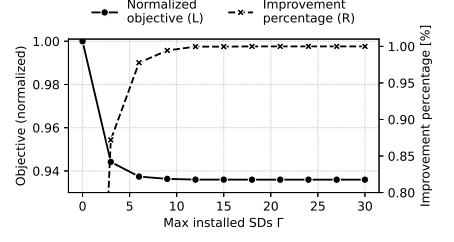
Effect of number of installed SDs Γ



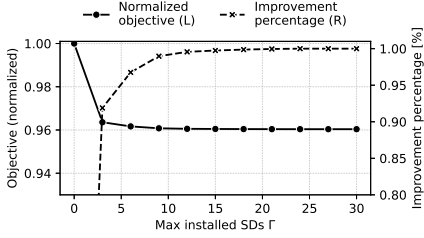
(a) Grid C0S1U2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



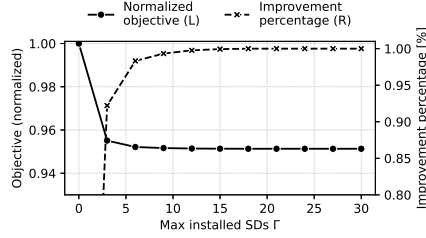
(b) Grid C1R0S2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



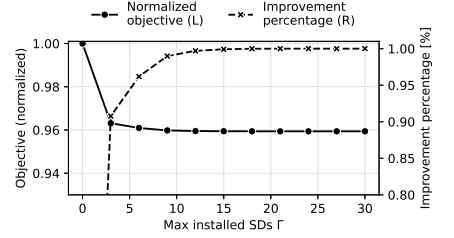
(c) Grid C2R1U0, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



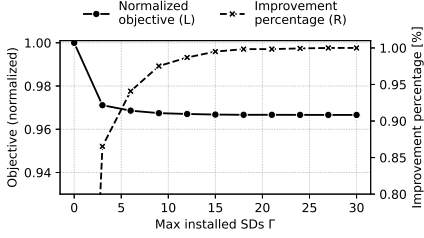
(d) Grid C0S1U2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



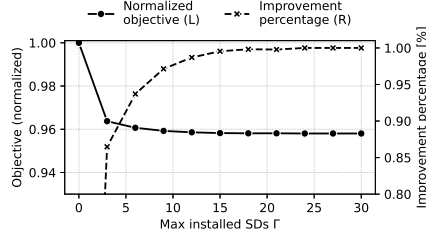
(e) Grid C1R0S2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



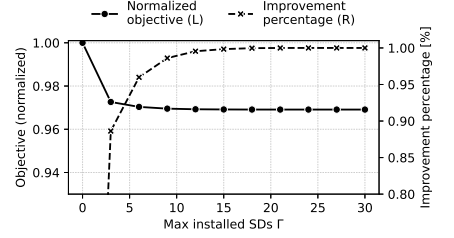
(f) Grid C2R1U0, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



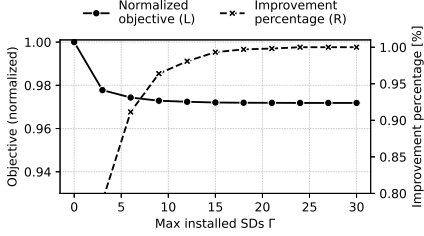
(g) Grid C0S1U2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



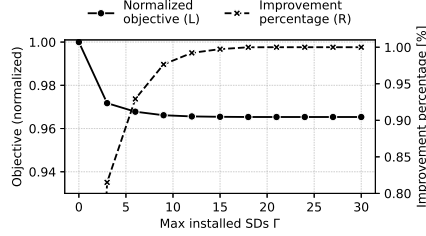
(h) Grid C1R0S2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



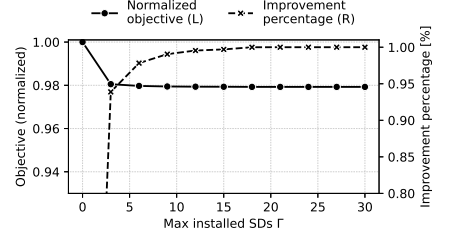
(i) Grid C2R1U0, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



(j) Grid C0S1U2, $\bar{P} = 0$, i.e. power flow minimization.



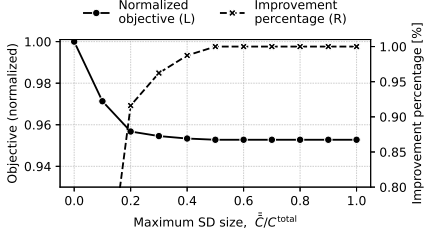
(k) Grid C1R0S2, $\bar{P} = 0$, i.e. power flow minimization.



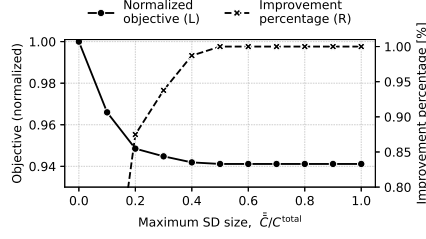
(l) Grid C2R1U0, $\bar{P} = 0$, i.e. power flow minimization.

Figure A.3: The effect of the number of installed SDs Γ for Virtual Power Line (VPL) operation, for various grids (columns) and congestion definitions (rows). For all figures, we observe a sharp early improvement followed by rapid saturation. Going from one SD per Medium Voltage MV-grid ($\Gamma = 3$) to two SDs per Medium Voltage MV-grid ($\Gamma = 6$) captures almost all of the attainable VPL benefit; beyond that, extra SDs give small marginal improvements.

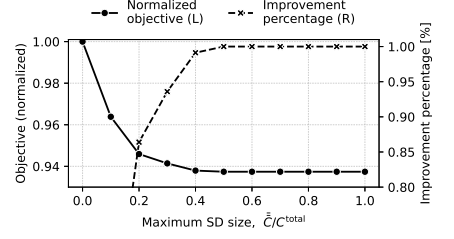
Effect of maximum SD size $\bar{C}$



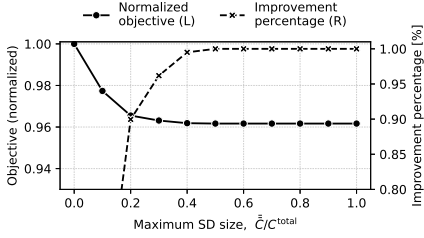
(a) Grid C0S1U2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



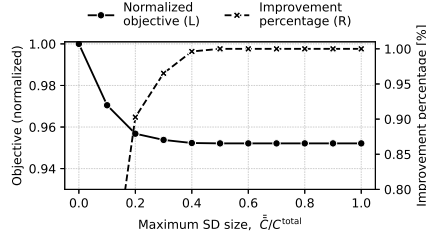
(b) Grid C1R0S2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



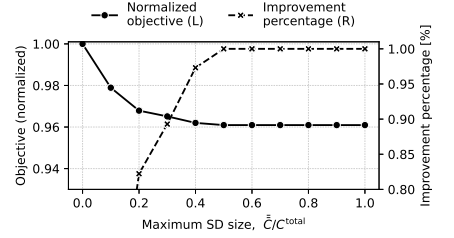
(c) Grid C2R1U0, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



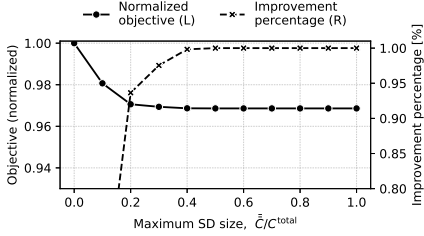
(d) Grid C0S1U2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



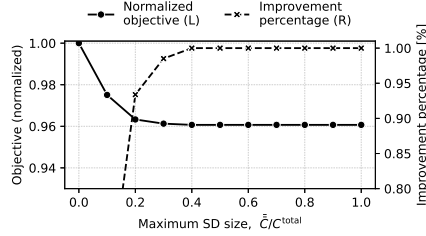
(e) Grid C1R0S2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



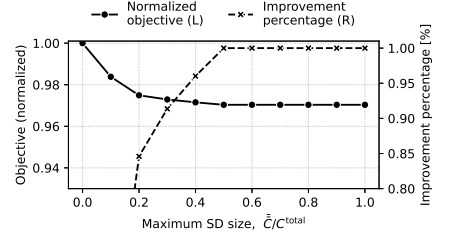
(f) Grid C2R1U0, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



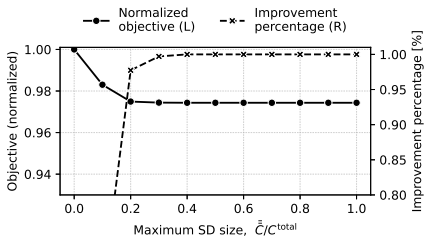
(g) Grid C0S1U2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



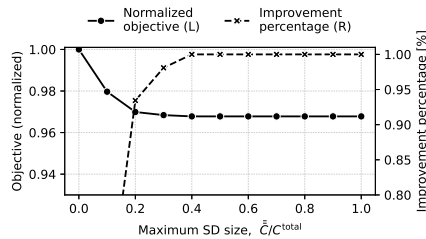
(h) Grid C1R0S2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



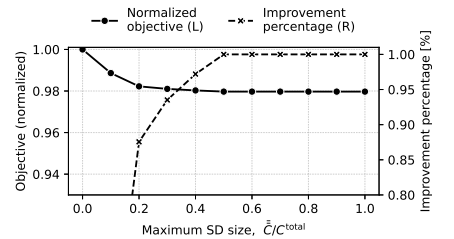
(i) Grid C2R1U0, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



(j) Grid C0S1U2, $\bar{P} = 0$, i.e. power flow minimization.



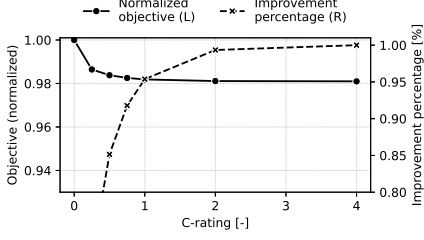
(k) Grid C1R0S2, $\bar{P} = 0$, i.e. power flow minimization.



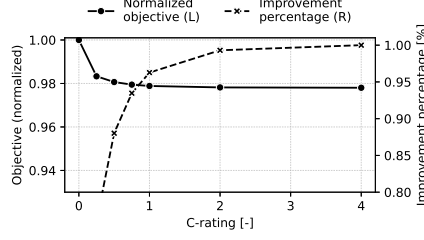
(l) Grid C2R1U0, $\bar{P} = 0$, i.e. power flow minimization.

Figure A.4: The effect of the maximum SD size $\bar{C}$, for various grids (columns) and congestion definitions (rows). In general, the figures tell a similar result: A large part of the possible reduction is obtained for reasonable low values of $\bar{C}$. Still, performance only saturates once $\bar{C} \approx 0.5C^{\text{total}}$, indicating that some nodes require large SDs. From Section ?? we know this is likely the slack node.

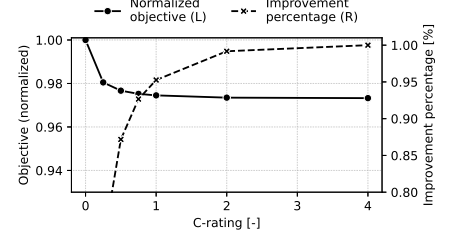
Effect of C-rating ρ



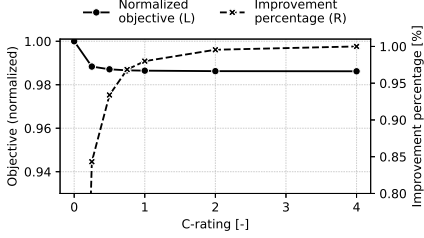
(a) Grid C0S1U2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



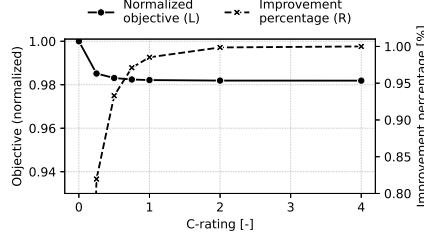
(b) Grid C1R0S2, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



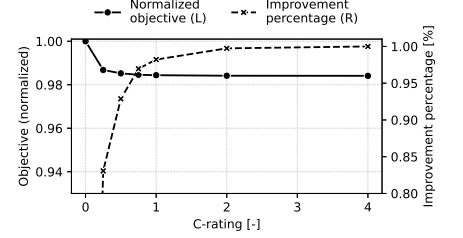
(c) Grid C2R1U0, $\bar{P}$ s.t. there are two cable types per subgrid, and there is congestion for 10% of time.



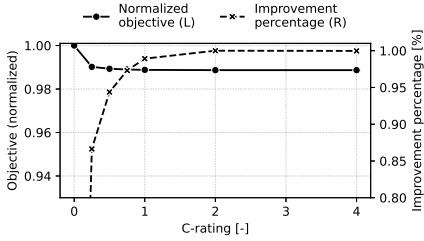
(d) Grid C0S1U2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



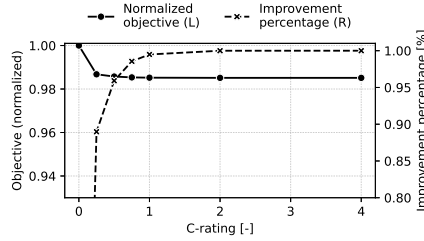
(e) Grid C1R0S2, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



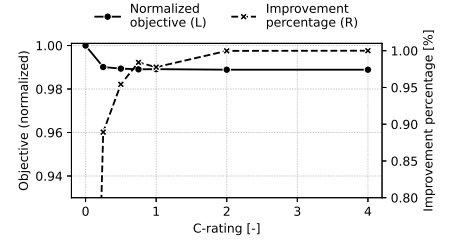
(f) Grid C2R1U0, $\bar{P}$ s.t. there are three cable types per subgrid, and there is congestion for 25% of time.



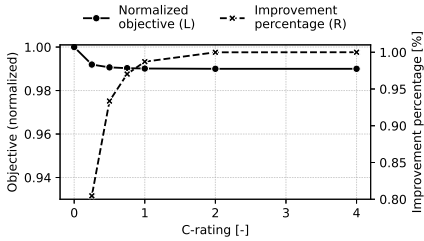
(g) Grid C0S1U2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



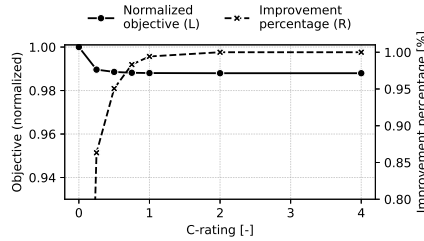
(h) Grid C1R0S2, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time.



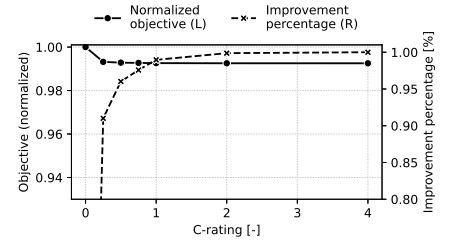
(i) Grid C2R1U0, $\bar{P}$ s.t. there are four cable types per subgrid, and there is congestion for 50% of time. The dip in improvement percentage around $\rho = 1$ is due to numerics.



(j) Grid C0S1U2, $\bar{P} = 0$, i.e. power flow minimization.



(k) Grid C1R0S2, $\bar{P} = 0$, i.e. power flow minimization.

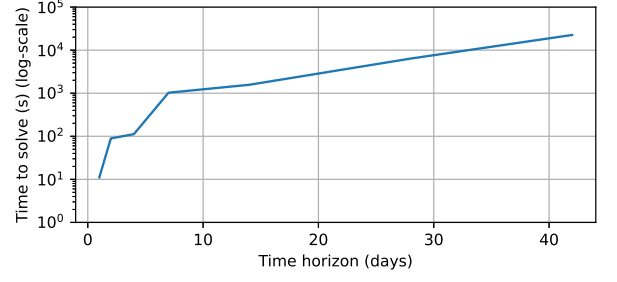


(l) Grid C2R1U0, $\bar{P} = 0$, i.e. power flow minimization.

Figure A.5: The effect of the C-rating ρ of installed SDs for Virtual Power Line (VPL) operation, for various grids (columns) and congestion definitions (rows). Overall we observe that a moderate C-rating of $\rho = 0.5$ obtains 90.5% of the available improvement (right y-axis).

II.1.1. Effect of time set size $|\mathcal{T}|$

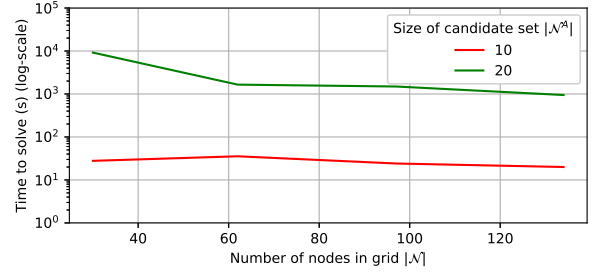
The results (Fig. B.6a) indicate that the solve time increases exponentially with the time set size, which is expected since a large part of the decision variables increase linearly with the number of time steps.



(a) Solve time for increasing time set size $|\mathcal{T}|$. The grid $\mathcal{G}$, the candidate set $\mathcal{N}^A$, and the number of SDs Γ is fixed.

II.1.2. Effect of grid size $|\mathcal{N}|$

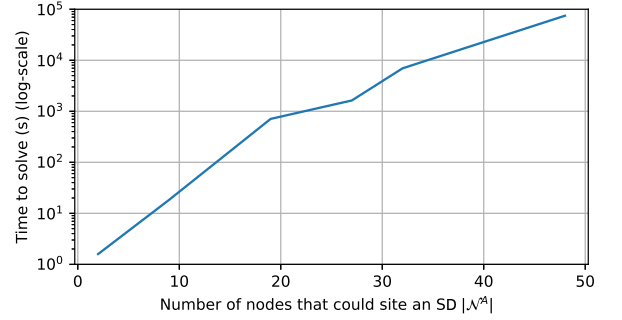
The results (Fig. B.6b) indicate that the solve time stays roughly constant or even decreases when increasing the number of nodes in the grid $|\mathcal{N}|$, for a fixed candidate set size $|\mathcal{N}^A|$. This is surprising, since this suggests that increasing the number of decision variables and parameters does not increase the complexity of the problem. The decrease might be caused by the fact that the nodes in the candidate set $\mathcal{N}^A$ are further apart from each other, and therefore the solver can identify more easily which nodes are most beneficial for the solution.



(b) Solve time for increasing grid size, for two candidate set sizes, $\mathcal{N}^A = \{10, 20\}$. The time set $\mathcal{T}$, and the number of SDs Γ is fixed.

II.1.3. Effect of candidate set $\mathcal{N}^A$

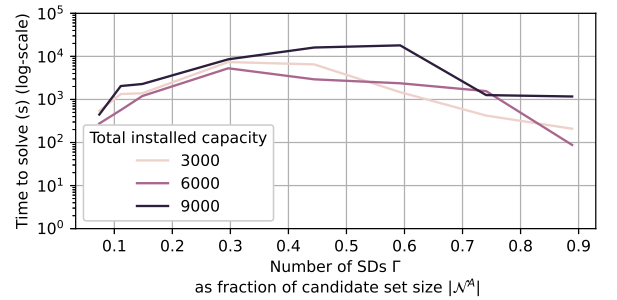
The results (Fig. B.6c) indicate that the solve time increases exponentially with the number of allowed locations $|\mathcal{N}^A|$. This is expected, since increasing the candidate set size increases the siting search space, which would (in a naive search) lead to a combinatorial increase in the number of options.



(c) Solve time for increasing size of candidate set $|\mathcal{N}^A|$. The grid $\mathcal{G}$, the time set $\mathcal{T}$, and the number of SDs Γ is fixed.

II.1.4. Effect of number of SDs Γ

The results (Fig. B.6d) indicate that the solve time is the largest when the number of SDs is half of the size of the candidate set $|\mathcal{N}^A|$, and that it decreases for larger values of Γ . This means the solve time resembles a binomial coefficient $\binom{|\mathcal{N}^A|}{\Gamma}$. This is somewhat surprising since for increasing Γ , the number of decision variables and constraints increases (there are more non-trivial charge-rates $\Delta_{t,i}$ to keep track of), but this appears to be offset by the fact that the solver effectively has a smaller search space during siting, when the number of SDs Γ approaches the candidate set size $|\mathcal{N}^A|$ (since $\binom{n}{n-1} < \binom{n}{n/2}, \forall n > 2$).



(d) Solve time for increasing number of allowed SDs Γ . The grid $\mathcal{G}$, the time set $\mathcal{T}$, and the candidate set $\mathcal{N}^A$ is fixed.

Figure B.6: Solve time measured in seconds (log scale) dependent on four different parameters: (a) Time set size $|\mathcal{T}|$ (b) Number of nodes $|\mathcal{N}|$. (c) Size of the candidate set $|\mathcal{N}^A|$. (d) Number of SDs Γ .

III. Numerical verification of DC-OPF validity

In this section we present an experiment to validate that DC-OPF does not implicitly optimize towards very high voltages or currents, since these are not explicitly captured in the DC-OPF formulation.

We do this by comparing the voltages and currents before and after Virtual Power Line (VPL) placement and operation. We determine the initial state using just the load profiles from the source dataset and use these to compute all voltages and current flows. The after state is determined by superimposing the optimal charge schedules Δ on initial load profiles $p \leftarrow p + \Delta$ and recomputing all voltages and current flows. In both cases we use reactive power $q = 0$. We compute the load flows using the `power-grid-model` Newton Raphson solver [?].

For all twelve scenarios we present the difference between the initial and final maximum voltage (Table C.1), minimum voltage (Table C.2), maximum current (Table C.3), and maximum power (Table C.4). We see a positive effect of the intervention for both current (Table C.3) and power (Table C.4). This indicates that the formulation optimizes towards sensible values. The effect on voltage is negligible, as all values in the voltage tables are 1.

Table C.1: Ratio of maximum node voltage v , before and after VPL intervention, computed using a Newton-Raphson load flow solver.

	$\bar{P}^{2,10\%}$	$\bar{P}^{3,25\%}$	$\bar{P}^{4,50\%}$	$\bar{P} = 0$
C0S1U2	1	1	1	1
C1R0S2	1	1	1	1
C2R1U0	1	1	1	1

Table C.2: Ratio of minimum node voltage v before and after VPL intervention, computed using a Newton-Raphson load flow solver.

	$\bar{P}^{2,10\%}$	$\bar{P}^{3,25\%}$	$\bar{P}^{4,50\%}$	$\bar{P} = 0$
C0S1U2	1	1	1	1
C1R0S2	1	1	1	1
C2R1U0	1	1	1	1

Table C.3: Ratio of maximum edge current I before and after VPL intervention, computed using a Newton-Raphson load flow solver.

	$\bar{P}^{2,10\%}$	$\bar{P}^{3,25\%}$	$\bar{P}^{4,50\%}$	$\bar{P} = 0$
C0S1U2	0.991	0.992	0.991	0.997
C1R0S2	1	1	1	1
C2R1U0	0.995	0.994	1	0.995

Table C.4: Ratio of maximum edge power P before and after VPL intervention, computed using a Newton-Raphson load flow solver.

	$\bar{P}^{2,10\%}$	$\bar{P}^{3,25\%}$	$\bar{P}^{4,50\%}$	$\bar{P} = 0$
C0S1U2	0.969	0.971	0.969	0.989
C1R0S2	0.875	0.929	0.966	0.962
C2R1U0	0.915	0.893	0.998	0.916

IV. Grid pre-processing Algorithms

This section describes how the grids were pre-processed. As a recap, first, three MV grid were optimized together

as a single grid group. To do this optimization, we merged the slack nodes in all MV grids to a single slack node. This is done using Alg. 1

Additionally, we reduced the size of the grids to reduce computational time. The algorithm is described in Alg. 2. Put briefly it works as follows: We look for so-called chains in the graph. A chain is a group of nodes that have exactly two neighbors, and at least one neighbor is also in the chain. Chains can be collapsed into a single node. We differentiate between 'bridges' (chains that end in nodes that have more than two neighbors) and 'appendages' (chains that have a member that has a neighbor which is a leaf node). For both bridges and appendages, the power consumption of the remaining node is equal to the sum of the chain nodes. For a bridge the impedance over the entire bridge is equal before and after the transformation. For appendages, the impedance between the resultant node and the rest of the graph is a discounted sum over all impedances in the appendage.

IV.1. Algorithm to merge slack nodes

Algorithm 1 Merge slack nodes

```

1: procedure MERGESLACKNODES( $\mathcal{G}$ )
2:    $n_s^* \leftarrow \emptyset$   $\triangleright$  Initialize target slack node as empty
3:   for all  $n \in \mathcal{N}^s$  do
4:     if  $n_s^* = \emptyset$  then  $\triangleright$  set the target slack node
5:        $n_s^* = n$ 
6:     else  $\triangleright$  Connect all edges connected to another
       slack node to the target slack node
7:       for all  $(i, j) \in \mathcal{E}$  do
8:         if  $i = n$  then
9:            $i = n_s^*$   $\triangleright$  replace redundant slack
           node  $n$  with target slack node  $n_s^*$ 
            $\triangleright$  Note that all edge attributes get transferred,
           e.g.  $r(n_s^*, j) = r(i, j)$ 
10:        end if
11:        if  $j = n$  then
12:           $j \leftarrow n_s^*$   $\triangleright$  idem
13:        end if
14:      end for
15:    end if
16:  end for
17:  return  $\mathcal{G}$ 
18: end procedure

```

IV.2. Reducing graph size by merging chains

Algorithm 2 Simplifying chains of a power grid into single nodes.

```

1: procedure SIMPLIFYGRID( $\mathcal{G}, p$ )
2:    $\mathcal{B} \leftarrow \{n \in \mathcal{N} \mid \text{degree}(n) \leq 2\}$ 
3:    $\mathcal{B} \leftarrow \text{sort } \mathcal{B} \text{ such that each chain is sorted in-chain.}$ 
4:   for all  $b = (0, \dots, k) \in \mathcal{B}$  with  $k \geq 1$  do
5:     if  $\text{degree}(0) = 1$  or  $\text{degree}(k) = 1$  then  $\triangleright$ 
       one endpoint is a leaf
6:        $\mathcal{G} \leftarrow \text{SIMPLIFYAPPENDAGE}(\mathcal{G}, p, b)$ 
7:       else if  $b = k$  then  $\triangleright$  loops are left unchanged
8:         continue
9:       else  $\triangleright$  bridge
10:         $\mathcal{G} \leftarrow \text{SIMPLIFYBRIDGE}(\mathcal{G}, p, b)$ 
11:      end if
12:    end for
13:  return  $\mathcal{G}$ 
14: end procedure

15: procedure SIMPLIFYAPPENDAGE( $\mathcal{G}, p, b$ )
16:   orient  $b$  so  $v_0$  is the grid-side end and  $v_k$  the leaf
17:    $a \leftarrow$  the node  $v_0$  is connected to that is not in  $b$ 
18:    $p_u \leftarrow \sum_{v \in b} p_v$ 
19:    $r_{(a, v_0)} \leftarrow r_{(a, v_0)} + \sum_{i=1}^k \frac{r_{(v_{i-1}, v_i)}}{i+1}$ ,  $x_{(a, v_0)} \leftarrow$ 
      $x_{(a, v_0)} + \sum_{i=1}^k \frac{x_{(v_{i-1}, v_i)}}{i+1}$   $\triangleright$  discounted series sum onto
     the surviving edge
20:    $i_{(a, v_0)} \leftarrow \min(i_{(a, v_0)}, \min_{1 \leq i \leq k} i_{(v_{i-1}, v_i)})$   $\triangleright$  keep
     the tightest current rating
21:   remove nodes  $v_1, \dots, v_k$  and edges
      $(v_0, v_1), \dots, (v_{k-1}, v_k)$ 
22:   return  $\mathcal{G}$ 
23: end procedure

24: procedure SIMPLIFYBRIDGE( $\mathcal{G}, p, b$ )
25:    $a \leftarrow$  the node  $v_0$  is connected to that is not in  $b$ 
26:    $z \leftarrow$  the node  $v_k$  is connected to that is not in  $b$ 
27:    $m \leftarrow$  pick any  $n \in b$  as the surviving node  $\triangleright$  the
     split below preserves  $a$ - $z$  impedance for any choice
28:    $p_u \leftarrow \sum_{v \in b} p_v$ 
29:    $R \leftarrow \sum_{i=1}^k r_{(v_{i-1}, v_i)}$ ,  $X \leftarrow \sum_{i=1}^k x_{(v_{i-1}, v_i)}$ ,  $I \leftarrow$ 
      $\min_{1 \leq i \leq k} i_{(v_{i-1}, v_i)}$   $\triangleright$  over the  $k$  interior edges of  $b$ 
30:   for all  $(g, v) \in \{(a, v_0), (z, v_k)\}$  do  $\triangleright$  the two
     edges linking  $b$  to the grid
31:      $r_{(g, v)} \leftarrow r_{(g, v)} + \frac{R}{2}$ ,  $x_{(g, v)} \leftarrow x_{(g, v)} + \frac{X}{2}$ ,
      $i_{(g, v)} \leftarrow \min(i_{(g, v)}, I)$ 
32:     reconnect  $(g, v) \rightarrow (g, m)$ 
33:   end for
34:   remove all  $v \in b \setminus \{m\}$  and edges
      $(v_0, v_1), \dots, (v_{k-1}, v_k)$ 
35:   return  $\mathcal{G}$ 
36: end procedure

```
